Stress Testing Factual Consistency Metrics for Long-Document Summarization

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Abstract

Evaluating the factual consistency of abstractive text summarization remains a significant challenge, particularly for long documents, where conventional metrics struggle with input length limitations and long-range dependencies. In this work, we systematically evaluate the reliability of six widely used reference-free factuality metrics, originally proposed for short-form summarization, in the long-document setting. We probe metric robustness through seven factuality-preserving perturbations applied to summaries, namely paraphrasing, simplification, synonym replacement, logically equivalent negations, vocabulary reduction, compression, and source text insertion, and further analyze their sensitivity to retrieval context and claim information density. Across three long-form benchmark datasets spanning science fiction, legal, and scientific domains, our results reveal that existing short-form metrics produce inconsistent scores for semantically equivalent summaries and exhibit declining reliability for information-dense claims whose content is semantically similar to many parts of the source document. While expanding the retrieval context improves stability in some domains, no metric consistently maintains factual alignment under long-context conditions. Finally, our results highlight concrete directions for improving factuality evaluation, including multi-span reasoning, context-aware calibration, and training on meaning-preserving variations to enhance robustness in long-form summarization.[1]

1 Introduction

Abstractive summarization has seen rapid advances with the advent of large language models (LLMs), but ensuring that generated summaries faithfully reflect the source content remains a persistent challenge (Laban et al., 2024; Wright et al., 2025). Summaries that read fluently can nonetheless introduce hallucinated details or omit critical facts (Belém et al., 2025), undermining their reliability for downstream tasks in domains such as medicine, law, and scientific review (Asai et al., 2024). Traditional evaluation measures like ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002), which rely on n-gram overlap with reference summaries, are useful for measuring surface similarity but fail to capture factual consistency, since two summaries can overlap heavily in wording while still differ in correctness (Maynez et al., 2020). This gap has motivated the development of reference-free factuality evaluation metrics, which assess whether the statements in a summary are supported by the source document itself rather than by comparison with a human-written reference, using techniques such as question answering (Wang et al., 2020; Scialom et al., 2021; Fabbri et al., 2022), natural language inference (Laban et al., 2022; Chen and Eger, 2023; Zha et al., 2023), or LLM-based scoring (Liu et al., 2023; Fu et al., 2024).

[1]We release all code, perturbed data, and scripts required to reproduce our results at https://github.com/zainmujahid/metricEval-longSum.

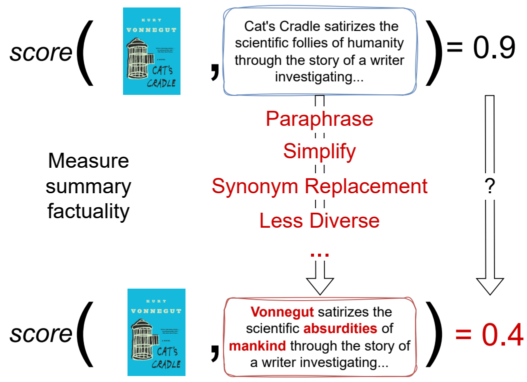

Figure 1: We aim to see how robust summary factuality metrics are for long and multi-document setups by applying meaning-preserving perturbations and comparing metric scores before and after these edits.

While such metrics have demonstrated promise on short-document datasets, their scalability to long-form summarization is likely to be hindered by challenges unique to long context lengths (Russak et al., 2024; Sarthi et al., 2024; Edge et al., 2024). Important details may be dispersed across hundreds or thousands of tokens and thus overlooked by metrics that process only truncated inputs (Laban et al., 2024); multi-document summaries must reconcile diverse writing styles and potentially conflicting information (Asai et al., 2024); and the reference-free setting deprives evaluators of gold annotations, necessitating robust intrinsic evaluation protocols. Our goal is to systematically benchmark widely used factual consistency metrics under these long-document conditions, in order to reveal their robustness and limitations.

Building on this foundation, we apply a stress-testing methodology established in previous work (Ramprasad and Wallace, 2024) to six widely-used reference-free metrics (§ 4) across seven factuality-preserving perturbations reflecting realistic long-form summarization phenomena. These perturbations (§ 3.1) broadly cover paraphrasing operations, simplification of complex constructions, synonym replacement, reduced lexical diversity, logically equivalent negations, further compression of the summary, and insertion of unrelated source sentences. They are designed to challenge metrics to distinguish genuine factual consistency from superficial cues and to maintain robustness in the face of lexical and structural variation. On top of this, we investigate the impact of long-document specific phenomena, including retrieval length and evidence dispersion (Goldman et al., 2024).

Through extensive experiments on six evaluation metrics across three benchmark datasets in science fiction, legal, and scientific domains, we expose significant weaknesses in existing metrics, such as inconsistent scoring across semantically equivalent summaries (Fig. 1). Our analysis reveals that current factuality metrics vary widely in robustness to meaning-preserving edits, with some highly sensitive to surface changes while others remain more stable. Most metrics benefit from broader retrieval context windows, though with notable domain-specific variation. We also find that metric reliability decreases for information-dense claims that overlap semantically with large portions of the source document, suggesting that current metrics struggle with compressed or globally entangled content. These insights point toward improving evaluation consistency by developing metrics that integrate multi-span reasoning and context-aware calibration.

2 Factual Consistency in Abstractive Summarization

Factual consistency refers to whether a summary accurately reflects the content of the source document. While abstractive summarization systems have become increasingly fluent, they often produce factually incorrect or hallucinated statements (Huang et al., 2025). These hallucinations can range from minor misstatements to major distortions, particularly when the input is lengthy and semantically dense (Belém et al., 2025). A number of techniques have been proposed to mitigate this (Gao et al., 2023; Qiu et al., 2023; Zhang et al., 2024; Mündler et al., 2024; Wang et al., 2024). Despite these efforts, most prior work has focused on short-form summarization settings, where the task is relatively controlled. In contrast, long-form summarization requires condensing information spread across thousands of tokens, making reliable factuality evaluation substantially more difficult, especially without reference summaries or human annotations.

2.1 Factuality Evaluation Metrics

To overcome the limitations of traditional reference-based metrics (Maynez et al., 2020), a range of reference-free metrics have emerged that assess the alignment between the summary and source directly. Entailment-based approaches such as FactCC (Kryscinski et al., 2020) and SummaC (Laban et al., 2022) use NLI models to judge whether summary sentences are supported by the source. QA-based methods like QAGS (Wang et al., 2020) and QuestEval (Scialom et al., 2021) evaluate factuality by generating and answering questions derived from the summary. Generation-based metrics, including `BARTScore` (Yuan et al., 2021) and T5Score (Trainin and Abend, 2025), estimate the likelihood of the summary given the source using pretrained sequence-to-sequence models. More recent tools like `AlignScore` (Zha et al., 2023), `MiniCheck` (Tang et al., 2024), and `UniEval` (Zhong et al., 2022) aim to improve efficiency and generalization across tasks. While effective on short-document benchmarks, most of these metrics assume the full source and summary can be jointly encoded, limiting their utility for long-form inputs. Moreover, recent work shows that these metrics are often brittle and sensitive to edits like paraphrasing, reordering, or logically equivalent reformulations (Ramprasad and Wallace, 2024). A recent survey highlights persistent limitations in robustness and long-document evaluation for factuality metrics (Lamsiyah et al., 2025). In this work, we study the behavior of six popular factuality metrics in long-document summarization using a retrieval-based scoring framework, and systematically evaluate their robustness to a set of controlled meaning-preserving perturbations.

2.2 Challenges in Long-Document Factuality Evaluation

Evaluating factual consistency in long documents introduces challenges that differ fundamentally from those in short texts. Long inputs often contain information that is dispersed, hierarchically structured, and cross-referential, requiring models to link evidence across distant sections or even multiple documents (Asai et al., 2024). This results in long-range dependencies and positional biases such as the "lost in the middle" effect (Liu et al., 2024). Yet, research into robust factuality metrics for long inputs remains limited. Koh et al. (2023) identified a clear gap in the literature for automatic evaluation methods tailored to long-document summarization. LongSciVerify (Bishop et al., 2024) and LongEval (Krishna et al., 2023) are two of the only available datasets with human factuality annotations in this setting. Chunk-based approaches like SMART (Amplayo et al., 2023), partially address this by sequentially processing document segments, but they remain computationally expensive and often inconsistent. A more scalable alternative is retrieval-based scoring, exemplified by LongDocFACTScore (Bishop et al., 2024), which retrieves top-k relevant source passages for each summary sentence, and computes sentence-level factuality scores that can be aggregated into a global metric. However, it remains unclear whether existing metrics, when used within such frameworks, behave consistently and robustly across varying retrieval configurations. Recent work suggests that uniformly aggregating sentence-level factuality scores can be suboptimal for long documents, and that discourse-aware aggregation can improve inconsistency detection (Zhong and Litman, 2025). Our study addresses this gap by analyzing these behaviors under controlled conditions and across multiple domains.

2.3 Adversarial Robustness of Metrics

Recent work has evaluated the robustness of factuality metrics by applying controlled perturbations to the summary or source (Goyal and Durrett, 2021; Chen et al., 2021; Gabriel et al., 2021). Ramprasad and Wallace (2024) showed that many metrics are brittle when faced with logically equivalent but lexically altered summaries, with even benign transformations such as reordering or simplification causing large score shifts. However, these evaluations have been primarily limited to short-document settings, where evidence is localized and both the source and summary can typically be processed jointly.

Extending this evaluation framework to long-document summarization is not a straightforward change of testbed. Long documents introduce additional challenges, including dispersed evidence, higher degrees of abstraction and compression, and the need for retrieval-based evaluation to overcome input length constraints. These factors fundamentally alter how factuality metrics operate and interact with the input. In this work, we therefore adopt the perturbation-based methodology of prior work as a foundation and systematically examine how these metrics behave under long-context conditions. Beyond this, we additionally analyze the effects of retrieval context size and claim information density, revealing new failure modes that arise specifically in long-document and multi-document settings. Our results show that brittleness observed in short documents persists and is often amplified in long-form summarization, motivating the need for factuality metrics that can reason over multi-span evidence rather than relying on local or surface-level cues.

3 Metric Robustness in Long-Form Summarization

To analyze the robustness of existing factuality metrics in long-form summarization, we evaluate six widely used reference-free metrics (§ 4) that span diverse architectures and scoring paradigms.

3.1 Perturbation Strategies

To evaluate the robustness of factuality metrics in a controlled manner, we apply meaning-preserving perturbations to the original summaries, as done in Ramprasad and Wallace (2024) in the short document case. These perturbations are designed to vary the summary's surface form (lexical choices,